

Quiz 1

NAME: _____

Q.1. [5 points] Simplify the following expression:

$$\sqrt[3]{3a^2b^3} \cdot \sqrt[3]{6a^5b}$$

We want to reduce the number of radicals present. We can start by combining these two radicals into a single radical over the whole product:

$$\sqrt[3]{3a^2b^3 \cdot 6a^5b}$$

Now inside the radical, we combine like with like (remember the product rule for exponential expressions) to get:

$$\sqrt[3]{18a^5b^4}$$

Lastly, we need to ask ourselves how many copies of a and b may be pulled out of the radical. Since we have a^5 , we can break this down into $a^3 \cdot a^2$, meaning we can pull one a out of the radical and an a^2 gets left behind. Similarly, we can pull one b out of the radical and a b gets left behind. So we have:

$$ab\sqrt[3]{18a^2b}$$

Now we just need to check if there are any factors of 18 that may be pulled out of the radical. 18 has a prime factorization of $2 \cdot 3^2$, but we see no cubed (or higher power) factors. So 18 remains in the radical, and our final answer is $ab\sqrt[3]{18a^2b}$.

Q.2. [5 points] Simplify the following expression:

$$\left[\frac{x-2y}{5z+1} \right] \div \left[\frac{3x-6y}{-z-(2+9z)} \right]$$

First let us simplify the denominator of the second fraction:

$$-z - (2 + 9z) = -z - 2 - 9z = -10z - 2$$

So our expression is:

$$\left[\frac{x-2y}{5z+1} \right] \div \left[\frac{3x-6y}{-10z-2} \right]$$

Converting division into multiplication:

$$\left[\frac{x-2y}{5z+1} \right] \cdot \left[\frac{-10z-2}{3x-6y} \right] = \frac{(x-2y) \cdot (-10z-2)}{(5z+1) \cdot (3x-6y)}$$

Now notice the common factors in the numerator and denominator. $3x - 6y = 3(x - 2y)$ and $-10z - 2 = -2(5z + 1)$. Cancelling these out we are left with:

$$\frac{-2}{3}$$

Q.3. [5 points] Fully factor the following polynomial:

$$x^2y + 3x^2 - 4y - 12$$

The GCF of the terms is just 1.

Since this polynomial has 4 terms, we should consider factoring by grouping. Indeed, the first two terms and the last two terms have common factors:

$$x^2(y + 3) - 4(y + 3)$$

and $y + 3$ is again a common factor:

$$(x^2 - 4)(y + 3)$$

But we're not done yet! The term $y + 3$ is a fully reduced "prime" polynomial factor. But $x^2 - 4$ is not. It is a difference of squares, and factors as $(x - 2)(x + 2)$. So our final answer is:

$$(x - 2)(x + 2)(y + 3)$$

Q.4. [5 points] Fully factor the following polynomial:

$$-3x^2 - 4x + 15$$

The GCF of the terms is just 1.

This is a trinomial (three terms) in one variable (namely, x). We should see if the p & q method works here.

Here $a = -3$, $b = -4$, and $c = 15$.

So $ac = -45$ and we need to find integers, p and q , such that $p + q = -4$ and $pq = -45$. The factorization of -45 that works is $(-9) \cdot 5 = -45$.

We now use these numbers, p and q , to split the middle term and then factor by grouping. $-4x = -9x + 5x$, so:

$$-3x^2 - 4x + 15 = -3x^2 - 9x + 5x + 15$$

$$= -3x(x + 3) + 5(x + 3)$$

$$= (-3x + 5)(x + 3)$$